# Exponential online mistakes at every width after arbitrarily small Gaussian anisotropy

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**Abstract**

We prove an exponential online-mistake lower bound for a Bayesian tanh network at every hidden width. The target is ordinary majority on $d$ odd input bits: it has a linear-size Boolean circuit and a width-one teacher with every weight at most one. The learner uses one hidden layer, a constant input coordinate, a linear readout, global absolute output margin greater than $1/2$, and exact training labels. The first-layer Gaussian variances have an arbitrarily small, explicitly specified coordinate asymmetry. One fixed input permutation forces at least $2^{\lfloor d/8 \rfloor}$ mistakes in the strong-prior limit, and forces all of them with arbitrarily high probability at sufficiently small positive variance. The covariance perturbation and input order are independent of the learner width. The proof bounds the energy of every nonlinear network using its essential input coordinates; it does not restrict the posterior to scalar subnetworks or choose an unfavorable classifier from a tie. For $d = 9q$ with odd $q$, a second construction uses only $2q$ calibration examples and forces $9^{(q+1)/2}$ mistakes, with an explicit inverse-polynomial upper bound on the anisotropy amplitude. The same priors also have width-independent polynomial IID sample complexity for this target. Neither construction establishes the same lower bound for the exactly isotropic Gaussian prior: at exact isotropy, the relevant candidate tier has equal masses and only a logarithmic cumulative sampling-error bound before it is exhausted.

## 1 Model and result

An earlier width-one anisotropic construction appears in [1]. The essential-coordinate bound below permits a comparison with every wide nonlinear competitor. It also yields a linear-calibration construction and a polynomial IID guarantee for the same prior and target.

Fix an odd input dimension $d \geq 17$ and a hidden width $N \geq 1$. Inputs lie in $\{-1, 1\}^d$. Write $\bar{x} = (1, x_1, \ldots, x_d)$ and consider the standard one-hidden-layer tanh network

$$z_{W,v}(x) = \sum_{a=1}^{N} v_a \tanh\left(\sum_{i=0}^{d} W_{ai}\bar{x}_i\right). \tag{1}$$

There are no biases other than the constant input coordinate. Choose a number $\delta \in (0, 1)$ and set

$$\lambda_0 = 1, \qquad \lambda_i = 1 + \delta 2^{-i} \quad (1 \leq i \leq d). \tag{2}$$

Independently, let

$$W_{ai} \sim N\left(0, \frac{N\sigma^2}{d\lambda_i}\right), \qquad v_a \sim N\left(0, \frac{\sigma^2}{N}\right). \tag{3}$$

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Its density is proportional to $\exp[-\kappa_{N,\lambda}/(2\sigma^2)]$, where

$$\kappa_{N,\lambda}(W,v) = \frac{d}{N}\sum_{i=0}^{d}\lambda_i\|W_{:,i}\|_2^2 + N\|v\|_2^2. \tag{4}$$

The parameter $\sigma$ is a common prior scale. Multiplying every variance in (3) by the same positive width-dependent factor merely reparametrizes this scale at a fixed width.

Before observing labels, condition the prior on

$$|z_{W,v}(x)| > \frac{1}{2} \qquad \text{for every } x \in \{-1,1\}^d. \tag{5}$$

After each prediction, reveal the true label and condition the posterior on its exact sign. The strong-prior limit is $\sigma \downarrow 0$ at fixed $d, N, \delta$. No width limit is taken in the proof.

For any desired $\eta > 0$, choosing $0 < \delta < \min\{1, \eta\}$ gives $1 \le \lambda_i \le 1 + \eta$ for every coordinate, uniformly in $d$ and $N$. The activation remains standard tanh. The model change is precisely the small coordinate dependence in (3); it is not the exactly isotropic first-layer law.

**Theorem 1.** *Let $d \ge 17$ be odd, put $k = \lfloor d/8 \rfloor$ and $q = 2k-1$, and use the target*

$$f(x) = \operatorname{sign}\left(\sum_{i=1}^{d} x_i\right).$$

*There is a fixed permutation of the input cube, depending only on $d$, with a designated set of $T$ rounds such that*

$$T \ge \frac{\binom{d}{k}}{\binom{2k-1}{k}} \ge 2^k. \tag{6}$$

*For every $N \ge 1$ and every $\delta \in (0,1)$, the posterior probability of the correct label on each designated round tends to zero as $\sigma \downarrow 0$ under (3)–(5).*

*Consequently, for every $\alpha \in (0,1)$ and every fixed $N, \delta$, all sufficiently small positive $\sigma$ have the following properties on that one fixed permutation:*

1. *Fresh posterior sampling makes all $T$ designated mistakes with probability at least $1-\alpha$.*
2. *Posterior-label majority is wrong on all $T$ designated rounds.*
3. *The strict $2/3$ confidence requirement fails on all $T$ designated rounds.*

*The target has bounded-weight teacher complexity $C_2(f) = 1$ and Boolean circuit size $\Theta(d)$ for a fixed bounded-fan-in complete gate basis. Thus every learner width $N \ge 1$ satisfies $N \ge C_2(f)$.*

The variance threshold may depend on $N, d, \delta, \alpha$. The theorem does not assert a positive threshold for this $\sigma$ uniform over all widths. Proposition 9 gives a width-uniform threshold in the mean-field scale $t = \sqrt{N}\sigma$. The covariance coefficients and input permutation themselves are the same for every width. The calibration prefix contains all inputs in two extreme Hamming regions; no short-calibration claim is made.

## 2 An energy bound for every nonlinear classifier

A coordinate $i \in \{1, \dots, d\}$ is *essential* to a Boolean function $g$ if some pair of inputs differing only in coordinate $i$ receives opposite labels. Let $\mathrm{ess}(g)$ be its set of essential input coordinates. A function depends only on this set: changing any other coordinate never changes its value.

Define

$$\Phi_d(Q) = \min_{u>0} \left\{ dQu^2 + \frac{1}{4}\coth^2 u \right\}, \qquad Q > 0. \tag{7}$$

For a represented function $g$, let $E_{N,\lambda}(g)$ be the infimum of (4) over parameters with $g(x)z_{W,v}(x) > 1/2$ at every cube input.

**Lemma 2** (Pair contraction)**.** *For real $c, t$,*

$$\frac{|\tanh(c+t) - \tanh(c-t)|}{2} \le \tanh|t|.$$

*The function $s \mapsto \tanh^2\sqrt{s}$ is concave on $[0,\infty)$. Consequently, for $w \in \mathbb{R}^N$,*

$$\sum_{a=1}^{N} \tanh^2|w_a| \le N\tanh^2\left(\frac{\|w\|_2}{\sqrt{N}}\right). \tag{8}$$

*Proof.* The identity

$$\tanh(c+t) - \tanh(c-t) = \frac{\sinh(2t)}{\cosh(c+t)\cosh(c-t)}$$

and $\cosh(c+t)\cosh(c-t) = (\cosh(2c) + \cosh(2t))/2$ show that the absolute difference is largest at $c = 0$. For concavity, put $t = \sqrt{s} > 0$. The derivative in $s$ is $\tanh(t)\operatorname{sech}^2(t)/t$; its derivative in $t$ has the sign of

$$t(1 - 3\tanh^2 t) - \tanh t < 0.$$

Indeed, $\tanh t \ge t\operatorname{sech}^2 t$, because $\sinh(2t) \ge 2t$, and substitution bounds the displayed expression by $-2t\tanh^2 t < 0$. Continuity handles $s = 0$. Jensen's inequality gives (8). □

**Lemma 3** (Weighted essential coordinates)**.** *For every nonconstant represented Boolean function $g$ and every width $N \ge 1$,*

$$E_{N,\lambda}(g) \ge \Phi_d\left(\sum_{i\in\mathrm{ess}(g)} \lambda_i\right). \tag{9}$$

*If $S \subseteq \{1, \dots, d\}$ has odd cardinality and $h_S(x) = \mathrm{sign}(\sum_{i\in S} x_i)$, then equality holds:*

$$E_{N,\lambda}(h_S) = \Phi_d\left(\sum_{i\in S} \lambda_i\right). \tag{10}$$

*The function $\Phi_d$ is strictly increasing.*

*Proof.* It suffices to work with weak signed margins at least $1/2$. The weak and strict infima agree: multiply the readout of a weakly feasible network by $1 + \rho$ and let $\rho \downarrow 0$.

Fix weakly feasible parameters and put $C = \sqrt{N}\|v\|_2$. For each essential $i$, choose adjacent inputs $x, x'$ with opposite labels. Their output difference has magnitude at least one. Writing the hidden preactivations as $c_a \pm W_{ai}$, applying Lemma 2, and using Cauchy–Schwarz gives

$$\begin{aligned}\frac{1}{2} &\le \frac{|z(x) - z(x')|}{2} \\ &\le \|v\|_2 \left( \sum_{a=1}^{N} \tanh^2 |W_{ai}| \right)^{1/2} \le C \tanh\left( \frac{\|W_{:,i}\|_2}{\sqrt{N}} \right).\end{aligned}$$

In particular $C > 1/2$. Set $u = \operatorname{arctanh}(1/(2C)) > 0$. Every essential column satisfies $\|W_{:,i}\|_2^2/N \ge u^2$. The bias and other columns contribute nonnegative energy, so

$$\kappa_{N,\lambda}(W, v) \ge d \left( \sum_{i \in \mathrm{ess}(g)} \lambda_i \right) u^2 + C^2 = d \left( \sum_{i \in \mathrm{ess}(g)} \lambda_i \right) u^2 + \frac{1}{4} \coth^2 u.$$

Minimizing proves (9).

For odd $|S|$, set every row of $W$ equal to $u$ on coordinates in $S$ and zero elsewhere, and set every $v_a = \coth(u)/(2N)$. The output is

$$z(x) = \frac{1}{2} \coth(u) \tanh\left( u \sum_{i \in S} x_i \right).$$

The integer sum has absolute value at least one, so the network has majority sign and weak margin at least $1/2$. Its energy is exactly the objective in (7) with $Q = \sum_{i \in S} \lambda_i$. Readout scaling and the lower bound prove (10).

For $Q > 0$, the objective in (7) tends to infinity at both endpoints $u \downarrow 0$ and $u \to \infty$, and therefore has a minimizer $u_Q > 0$. If $Q_2 > Q_1$, then

$$\Phi_d(Q_2) \ge \Phi_d(Q_1) + d(Q_2 - Q_1) u_{Q_2}^2 > \Phi_d(Q_1).$$

□

The lower bound applies to every wide nonlinear network. In particular, it supplies a comparison with functions outside the majority family, which cannot be obtained by examining scalar candidate subnetworks alone.

## 3 Calibration and an exact minimum-energy tier

Write $\mathrm{wt}(x) = |\{i : x_i = +1\}|$ and put $q = 2k - 1$. Reveal every input satisfying $\mathrm{wt}(x) \le k - 1$ with label $-1$, and every input satisfying $\mathrm{wt}(x) \ge d - k + 1$ with label $+1$. Use any fixed order within this prefix. All its labels agree with the full majority target because $k < d/2$.

**Lemma 4** (Two-sided calibration)**.** *Every Boolean function consistent with the calibration prefix has at least $q$ essential input coordinates. If it has exactly $q$, then it is $h_S$ for its essential support $S$, with $|S| = q$. Every such majority $h_S$ is consistent with the prefix.*

*Proof.* Suppose a consistent function depends on a support of size $r \le 2k - 2$. Choose an assignment on that support with $t$ positive coordinates, where

$$\max\{0, r - k + 1\} \le t \le \min\{r, k - 1\}.$$

This interval is nonempty. Setting all outside coordinates negative gives a negative calibration input, while setting all of them positive gives a positive calibration input, since $t + d - r \geq d - k + 1$. The two inputs agree on the support, a contradiction.

If $r = q = 2k - 1$, any support assignment with at most $k - 1$ positives extends to a negative calibration input by setting outside bits negative. Any assignment with at least $k$ positives extends to a positive calibration input by setting outside bits positive, because $k + d - q = d - k + 1$. These cases exhaust the support cube and force majority. Conversely, a $q$-coordinate majority is negative when the full input has at most $k - 1$ positives, and positive when the full input has at most $k - 1$ negatives. □

**Proposition 5** (Uniform separation of all competitors)**.** *At every width $N$, all functions $h_S$ with $|S| = q$ have pairwise distinct energies, and each is strictly cheaper than every other initially consistent Boolean function. After any further labels, while at least one of these candidates survives, the unique minimum-energy consistent function is the surviving $h_S$ with smallest $\sum_{i \in S} 2^{-i}$.*

*Proof.* By Lemma 3, the candidate energy is exactly

$$\Phi_d(Q_S), \qquad Q_S = \sum_{i \in S} \lambda_i = q + \delta \sum_{i \in S} 2^{-i}.$$

Finite binary subset sums are distinct. For example, at the smallest index belonging to exactly one of two subsets, its contribution $2^{-i}$ is larger than the sum of all later contributions through index $d$. Strict monotonicity of $\Phi_d$ therefore separates the candidate energies. Also

$$Q_S < q + \delta < q + 1. \tag{11}$$

Every outsider has at least $q + 1$ essential coordinates by Lemma 4, and every $\lambda_i \geq 1$. Its energy is at least $\Phi_d(q + 1)$, strictly above every candidate energy by (11). The same comparison holds after further conditioning whenever a candidate remains. □

Both the separation from outsiders and the ordering within the candidate family are independent of $N$. The ordering within the family is also independent of the positive value of $\delta$.

## 4 The input order and posterior mistakes

After calibration, repeat the following deterministic procedure while some candidate remains. Let $S$ be the surviving $q$-set with smallest $\sum_{i \in S} 2^{-i}$. Choose its lexicographically first $k$-element subset $A$ and query $x^A$, the vector positive exactly on $A$. Its true majority label is negative because $k < d/2$. The selected candidate satisfies $h_S(x^A) = +1$.

This query is new: any previously queried negative input would have eliminated a candidate that predicted positively there. It is outside the calibration prefix because it has weight exactly $k$. It eliminates exactly those candidate supports containing $A$. Indeed, a majority on $2k - 1$ coordinates predicts positively at a weight-$k$ input precisely when all its $k$ positive coordinates lie in that support.

There are initially $\binom{d}{2k-1}$ candidates. Each query eliminates at most $\binom{d-k}{k-1}$ of them, so exhausting the candidate class requires at least

$$T \geq \frac{\binom{d}{2k-1}}{\binom{d-k}{k-1}} = \frac{\binom{d}{k}}{\binom{2k-1}{k}} \geq \frac{(d/k)^k}{4^k} \geq 2^k.$$

Here $\binom{d}{k} \geq (d/k)^k$, $\binom{2k-1}{k} \leq 4^k$, and $d \geq 8k$. The entire sequence is computable before the run. It depends only on the candidate supports and the fixed negative labels, not on posterior samples. Complete it to a permutation by revealing all remaining cube inputs in a fixed order with their majority labels.

**Lemma 6** (Gaussian concentration on a unique minimum)**.** *At a fixed finite width and after any fixed feasible label history, if a unique consistent Boolean function has the least value of $E_{N,\lambda}$, then its posterior probability tends to one as $\sigma \downarrow 0$.*

*Proof.* For a represented Boolean function $g$, write

$$U_g = \{(W, v) : g(x) z_{W,v}(x) > 1/2 \text{ for every } x\}.$$

This is open. Its energy infimum agrees with that over its closed weak-margin set, by readout scaling. The latter set is nonempty and the positive definite quadratic energy is coercive, so the weak infimum is attained. In the finite-dimensional Gaussian parameter space,

$$\lim_{\sigma \downarrow 0} \sigma^2 \log \mathbb{P}_\sigma(U_g) = -\frac{1}{2} E_{N,\lambda}(g). \tag{12}$$

For an elementary lower bound, take a small ball contained in $U_g$ whose maximum energy is arbitrarily close to its infimum; Gaussian normalizing powers of $\sigma$ disappear after multiplication by $\sigma^2$. For the upper bound, write the energy as $(1-\varepsilon)\kappa + \varepsilon\kappa$, bound the first part below by $(1-\varepsilon)E_{N,\lambda}(g)$, and integrate the second Gaussian factor over the entire parameter space. The resulting constant $\varepsilon^{-P/2}$, where $P$ is the parameter dimension, also disappears at the logarithmic scale. Then let $\varepsilon \downarrow 0$.

There are finitely many Boolean functions on the cube. The globally margin-conditioned posterior after a label history is the prior restricted to the union of the compatible events $U_g$. A unique minimum energy has a positive gap from all other represented compatible functions. Equation (12) makes the ratio of each other event's probability to that of the minimum tend to zero. □

By Proposition 5, every designated query is predicted positively by the unique minimum-energy consistent function, while its true label is negative. Lemma 6 proves the claimed correct-label probability limit. For fixed $N, \delta$ there are only finitely many designated rounds, so one sufficiently small positive variance makes the correct-label probability smaller than $\min\{\alpha/T, 1/3\}$ on each. A union bound shows that all designated posterior samples are wrong with probability at least $1-\alpha$. The majority and confidence conclusions follow from the same bound.

**Target resources.** The width-one teacher $z(x) = \tanh(\sum_i x_i)$ has every weight at most one, and its signed margin is at least $\tanh(1) > 1/2$ because $d$ is odd. Thus $C_2(f) = 1$. For a linear-size Boolean circuit, convert the input bits to zero-one form and count them with carry-save adders. Each constant-size full adder replaces three equally weighted bits by two, decreasing the number of stored bits by one. At most $O(d)$ such adders leave two $O(\log d)$-bit numbers; their final addition and comparison with $d/2$ use $O(\log d)$ further gates. This gives an $O(d)$-gate circuit. Every input is essential to majority, so bounded fan-in gives an $\Omega(d)$ gate lower bound. This completes the proof of Theorem 1.

## 5 A linear calibration prefix

The preceding result permits every fixed $\delta \in (0, 1)$, but its calibration is large. A compactness argument gives a shorter prefix when the perturbation is sufficiently small for each dimension. The

quantitative bound below is explicit; we first give the shorter compactness argument explaining why a width-independent gap exists.

**Theorem 7** (Linear calibration)**.** *Let $q \geq 3$ be odd and $d = 9q$. For every*

$$0 < \delta \leq \frac{1}{80000q^3}, \tag{13}$$

*the model (1)–(5), with the same full-majority target, has a fixed input permutation beginning with exactly $2q$ calibration inputs and followed by exactly*

$$T = 9^{(q+1)/2} = \exp(\Omega(d)) \tag{14}$$

*designated mistake rounds. For every width $N \geq 1$, the correct-label posterior probability on each designated round tends to zero as $\sigma \downarrow 0$. All finite-positive-variance conclusions of Theorem 1 follow as well.*

*The chosen $\delta$ and input permutation are independent of $N$. The perturbation can consequently be arbitrarily small. Its explicit upper bound depends on $d$; no positive dimension-uniform amplitude is asserted in this linear-calibration theorem.*

### 5.1 A uniform gap above the majority minimum

The following relaxation is used only to prove an energy gap uniform over all widths. It does not replace the finite-width Gaussian posterior. For a probability measure $\nu$ on $\mathbb{R}^{d+1} \times \mathbb{R}$, write its variables as $(a, b)$ and define

$$z_\nu(x) = \int b \tanh(a \cdot \bar{x})\, d\nu(a,b), \qquad I(\nu) = \int (d\|a\|_2^2 + b^2)\, d\nu(a,b). \tag{15}$$

Every width-$N$ network is included: take $\nu = N^{-1} \sum_{j=1}^{N} \delta_{(W_{j,:}, N v_j)}$. Then $z_\nu = z_{W,v}$ and $I(\nu) = \kappa_{N,\mathbf{1}}(W, v)$.

**Lemma 8** (Compactness and equality)**.** *For fixed $d$ and odd $q \leq d$, there is $\Delta = \Delta(d, q) > 0$ such that every globally weak-margin classifier represented by (15) and having at least $q$ essential input coordinates either is a signed majority on exactly $q$ coordinates or has energy at least $\Phi_d(q) + \Delta$.*

*Proof.* First, energy sublevels are sequentially compact for the weak topology of probability measures. The second-moment bound gives tightness, and the integral of the nonnegative continuous quadratic energy is lower semicontinuous. For each fixed cube input, $z_\nu(x)$ is continuous along an energy-bounded weakly convergent sequence. Indeed, $|b \tanh(a \cdot \bar{x})| \leq |b|$ and

$$\int |b| \mathbf{1}_{\{|b| > R\}}\, d\nu \leq \frac{I(\nu)}{R}.$$

Truncating the $b$ coordinate gives bounded continuous integrands and the displayed uniform tail bound removes the truncation. Consequently, for any fixed Boolean function, its minimum weak-margin energy over measures is attained whenever its feasible set contains a measure of finite energy. This uses only tightness of probability measures on a finite-dimensional Euclidean space and the finite number of margin constraints.

The essential-coordinate proof of Lemma 3 applies with integrals in place of empirical averages. If the essential support has size $r$, it gives $I(\nu) \geq \Phi_d(r)$. We examine equality for $r = q$ at $I(\nu) = \Phi_d(q)$.

Put $C = (\int b^2\, d\nu)^{1/2}$ and $u = \operatorname{arctanh}(1/(2C)) > 0$. Equality in the summed energy bound forces $\int a_j^2\, d\nu = u^2$ for every essential $j$, and all unused coordinates, including $a_0$, to vanish almost surely. For each essential $j$, fix an adjacent pair with opposite labels. All inequalities in the pair-contraction, Jensen, and Cauchy–Schwarz chain must be equalities; otherwise its column-norm bound is strict and the total energy exceeds $\Phi_d(q)$.

Strict concavity in Lemma 2 gives $|a_j| = u$ almost surely. Equality in the scalar pair contraction forces the center of the two preactivations to be zero, because $u > 0$. Equality in Cauchy–Schwarz then gives, for a sign $\gamma_j \in \{-1, 1\}$ fixed by the orientation of the chosen pair,

$$b = \gamma_j C \operatorname{sign}(a_j) \quad \text{almost surely.}$$

Thus $|b| = C$ and $a_j = (u/C)\gamma_j b$ for every essential $j$. Intersecting these finitely many full-measure events yields

$$z_\nu(x) = C \tanh\left(u \sum_{j \in S} \gamma_j x_j\right).$$

Its Boolean sign is a signed $q$-coordinate majority. This proves that no other function with exactly $q$ essential coordinates can attain energy $\Phi_d(q)$.

There are finitely many Boolean functions on the cube. Attainment therefore gives a common positive gap to all finite-energy, non-majority functions with exactly $q$ essential coordinates. Functions with more than $q$ essential coordinates have the positive gap $\Phi_d(q+1) - \Phi_d(q)$. The minimum of these finitely many positive gaps is a suitable $\Delta$. Functions with no finite-energy representation need not be included in the minimum. □

### 5.2 Block calibration and elimination

Partition the $d = 9q$ coordinates into $q$ blocks $B_1, \ldots, B_q$ of size nine, with block indices read cyclically. Put $m = (q-1)/2$. For each $j$, use two block-constant calibration inputs. In the positive one, the positive blocks are $\{j, j+1, \ldots, j+m\}$; in the negative one, remove block $j$ from this set. The inputs in each pair differ only on block $B_j$, and the other $q-1$ blocks are balanced. The resulting $2q$ inputs are distinct and have the indicated full-majority labels.

A consistent Boolean function must have an essential coordinate in every block, since otherwise it would give equal labels to that block's two calibration inputs. In particular it has at least $q$ essential coordinates. A signed majority on exactly $q$ coordinates must use one coordinate per block. Its sign on block $j$ must be positive: a negative coefficient would make the output score decrease when that block changes from the negative calibration input to the positive one, contradicting the prescribed labels. Conversely, every positive majority with one coordinate in each block fits the calibration. Let $\mathcal{C}$ be these $9^q$ candidates.

Choose $\Delta$ from Lemma 8. If $u_q > 0$ minimizes (7) at $Q = q$, then

$$\Phi_d(q+\delta) \le \Phi_d(q) + d\delta u_q^2.$$

For the qualitative argument, choose $0 < \delta < \min\{1, \Delta/(2du_q^2)\}$. Appendix A supplies $\Delta_q = 1/(10000q^3)$ and $du_q^2 \le 4$, which justify every $\delta$ in (13): candidate energies are then strictly below $\Phi_d(q)+\Delta_q/2$, while outsiders have energy at least $\Phi_d(q)+\Delta_q$. For the explicit conclusion, take $\Delta = \Delta_q$ in the following comparison. Every candidate has exact anisotropic energy $\Phi_d(q + \delta\sum_{i\in S} 2^{-i}) < \Phi_d(q+\delta)$. Every initially consistent outsider has isotropic energy at least $\Phi_d(q) + \Delta$, and its anisotropic energy is no smaller because $\lambda_i \ge 1$. Thus every candidate is cheaper than every outsider, uniformly over all finite widths. Distinct binary subset sums give a unique cheapest surviving candidate at every stage.

Put $\ell = (q+1)/2$ and fix the first $\ell$ blocks. If $S$ is the current cheapest surviving support, query the input positive only at its chosen coordinates in those fixed blocks:

$$A = S \cap (B_1 \cup \cdots \cup B_\ell), \qquad x = x^A.$$

The query has $\ell$ positive coordinates, so its full-majority target label is negative, while the selected candidate predicts positively. A candidate $h_{S'}$ is positive there exactly when $S'$ contains all $\ell$ coordinates of $A$: it needs at least $\ell$ positive coordinates to win its $q$-coordinate majority, and the query has only $\ell$ positives in total.

Thus each query deletes precisely one complete projection fiber, consisting of the $9^{q-\ell}$ supports with that choice of one coordinate in each of the fixed $\ell$ blocks. There are $9^\ell$ such fibers. Each selected candidate belongs to a surviving fiber, so the procedure visits every fiber once and takes exactly $T = 9^\ell$ forced queries. This proves (14). No query repeats, and these queries lie outside the calibration prefix: their weight is $\ell$, whereas calibration weights are $9(q-1)/2$ and $9(q+1)/2$.

The procedure uses only the fixed candidate order and previous target labels, so it defines one fixed input sequence, completed arbitrarily to a permutation. The Gaussian concentration and finite-variance arguments from Lemma 6 apply at every fixed width without change. This proves Theorem 7.

## 6 A common mean-field scale at every width

The preceding theorems use the scale $\sigma$ in (3). There is also a natural parametrization with a single positive scale that works uniformly over all widths. Put $t = \sqrt{N}\sigma$, so that

$$W_{ai} \sim N\left(0, \frac{t^2}{d\lambda_i}\right), \qquad v_a \sim N\left(0, \frac{t^2}{N^2}\right). \tag{16}$$

This is the same family of priors with a width-dependent change of scale variable.

**Proposition 9** (Uniform small mean-field scale)**.** *Fix either construction above, including its allowed positive anisotropy and its designated $T$ rounds. For every $\alpha \in (0,1)$ there is $t_0 > 0$, independent of $N$, such that for every $N \geq 1$ and every $0 < t < t_0$, the posterior under (16) makes all $T$ designated sampling mistakes with probability at least $1-\alpha$. The posterior-majority and confidence conclusions are uniform in $N$ as well.*

*Proof.* At a designated round, let $h_S$ be the unique minimum-energy consistent function and let $E = \Phi_d(Q_S)$. The proofs above give a number $g > 0$, independent of $N$, such that every consistent function predicting the next label correctly has energy at least $E+g$. For the large-prefix construction this follows from the finitely many explicit candidate gaps and the outsider bound $\Phi_d(q+1)$. For the short-prefix construction use the uniform outsider gap from Lemma 8 and the same explicit candidate gaps.

Write $a_a = W_{a,:}$ and $b_a = Nv_a$. Under (16), the pairs $(a_a, b_a)$ are IID Gaussian, with coordinate variances $t^2/(d\lambda_i)$ and $t^2$. Let

$$J(a,b) = d\sum_{i=0}^{d} \lambda_i a_i^2 + b^2, \qquad P = d+2.$$

Then $\kappa_{N,\lambda} = N^{-1}\sum_a J(a_a, b_a)$ and $J(a_a, b_a)/t^2$ is chi-squared with $P$ degrees of freedom.

A cloned majority minimizer is a single pair $(a,b)$ of cost $E$. Slightly enlarge its readout and take a sufficiently small open ball $B$ about the resulting pair. Every pair in $B$ has the same strict

signed margins as $h_S$, and $\sup_B J \le E + g/4$. If all $N$ pairs lie in $B$, their averaged output has those strict margins and fits the current history. Its prior probability is $p_t^N$, where Gaussian density integration gives

$$p_t \ge c_B t^{-P} \exp\left(-\frac{E + g/4}{2t^2}\right)$$

for a positive constant $c_B$ independent of $N, t$.

Set $\zeta = g/[4(E + g + 1)] \in (0, 1)$. The chi-squared moment-generating function gives

$$\mathbb{P}_t\{\kappa_{N,\lambda} \ge E + g\} \le \left[\zeta^{-P/2} \exp\left(-\frac{(1-\zeta)(E+g)}{2t^2}\right)\right]^N.$$

The numerator of the posterior correct-label probability is bounded by this tail; its denominator is at least $p_t^N$. Since $\zeta(E + g) \le g/4$, their ratio is at most

$$\left[c_B^{-1} \zeta^{-P/2} t^P \exp\left(-\frac{g}{4t^2}\right)\right]^N = r(t)^N.$$

Here $r(t) \to 0$ as $t \downarrow 0$. Once $r(t) < 1$, this bound is at most $r(t)$ for every $N \ge 1$. There are finitely many designated rounds, so choose a common $t_0$ making every bound smaller than $\min\{\alpha/T, 1/3\}$. A union bound finishes the proof. □

## 7 Polynomial IID learning for the same prior

The online lower bounds do not arise from a lack of polynomial IID sample guarantees for this target. The following bound is independent of the learner width and applies to both constructions.

**Theorem 10** (IID upper bound)**.** *Let $d$ be odd, $0 < \delta < 1$, and use the majority target and prior (2)–(5). Define*

$$K = 5d + 1, \qquad m = \lceil 32K((d+1)\log 2 + 1)\rceil, \qquad B_d = m((d+1)^2 + 1)\log 2.$$

*For every width $N \ge 1$, every input distribution $\mu$, and every $\rho, \beta \in (0, 1)$, if*

$$n \ge \frac{6}{\rho}\left(B_d + \log\frac{1}{\beta}\right), \tag{17}$$

*then, at all sufficiently small positive prior scales, with probability at least $1 - \beta$ over $n$ IID labeled examples, the $\mu$-mass of inputs whose posterior correct-label probability is at most $2/3$ is at most $\rho$. The sample bound is $O((d^4 + \log(1/\beta))/\rho)$. For fixed $N, d, \delta, \rho$, the sufficiently-small-scale threshold can be chosen uniformly over all target-consistent histories and sample sizes. In the mean-field parametrization (16), one sufficiently small positive $t$ works uniformly over all widths as well.*

*Proof.* We first give an energy-bounded realization of the target. Put $a = d^{-1/2}$, give every hidden row coefficient $a$ on every input coordinate and zero bias, and give every readout coefficient $1/(N \tanh a)$. Its output is

$$z(x) = \frac{\tanh(a \sum_i x_i)}{\tanh a},$$

which has signed margin at least one. Its energy is

$$K_0 = \sum_{i=1}^{d} \lambda_i + \coth^2 a < d + 1 + 4d = K,$$

using $\sum_i \lambda_i < d+1$ and $\tanh a \geq a/2$ for $0 < a \leq 1$.

Next, every width-$N$ network with energy at most $K$ satisfies $\sum_a |v_a| \leq \sqrt{N}\|v\|_2 \leq \sqrt{K}$. If $U$ is uniform on $(-1,1)$, then

$$\tanh t = \mathbb{E}\,\operatorname{sign}\bigl(t - \operatorname{arctanh} U\bigr).$$

Choose neuron $a$ with probability $|v_a|/\sqrt{K}$, draw $U$, and output the signed affine threshold

$$h(x) = \operatorname{sign}(v_a)\operatorname{sign}\bigl(W_{a,:}\cdot \bar{x} - \operatorname{arctanh} U\bigr).$$

Use the zero function with the remaining probability. Then $h(x) \in \{-1,0,1\}$ and $\mathbb{E}h(x) = z(x)/\sqrt{K}$. For $m$ independent voters, Hoeffding's inequality and a union bound over the cube give

$$\mathbb{P}\left\{\max_x \left|\frac{\sqrt{K}}{m}\sum_{j=1}^{m} h_j(x) - z(x)\right| \geq \frac{1}{4}\right\} \leq 2^{d+1}e^{-m/(32K)} < 1.$$

Thus a globally weak-margin network of energy at most $K$ has the same sign as some sum of $m$ such voters. Almost surely each nonzero voter is a strict affine threshold on the finite cube, and its negative is again a strict affine threshold. The approximating sum has no zeros because the network's absolute margin is at least $1/2$.

There are at most

$$H_d \leq 2\sum_{j=0}^{d}\binom{2^d-1}{j} \leq 2^{(d+1)^2}$$

strict affine threshold functions on the cube, by the central hyperplane arrangement bound in coefficient dimension $d+1$. Allowing the zero voter, the class $\mathcal{H}_K$ of all resulting sign functions has size at most $(H_d+1)^m \leq e^{B_d}$. Every globally margin-separated energy-$K$ function belongs to this one finite class, uniformly over all learner widths.

The posterior mass of parameters with energy greater than $K$ tends to zero uniformly over target-consistent histories. To see this, let $U_f$ be the target's strict global-margin event. It contains an open ball of energy strictly below $K$, by the witness above. Every target-consistent posterior conditioning event contains $U_f$, so, using probabilities under the unconditioned Gaussian prior,

$$\mathbb{P}_\sigma\{\kappa > K \mid \text{global margin and history}\} \leq \frac{\mathbb{P}_\sigma\{\kappa > K\}}{\mathbb{P}_\sigma(U_f)}.$$

The Gaussian tail in the numerator has rate at least $K/2$ and the denominator has a strictly smaller rate, by the open ball. Hence the ratio tends to zero at each fixed width. It is independent of the history and sample size. Choose the scale small enough to make it at most $\rho/6$. In the mean-field parametrization, use the product-ball and chi-squared estimate from Proposition 9 with $E = K_0$ and $g = K - K_0$; the strict target witness supplies the denominator ball. This makes the same tail bound uniform over every width, in addition to every history.

For a fixed $g \in \mathcal{H}_K$ of $\mu$-error greater than $\rho/6$, the probability that it fits $n$ IID examples is at most $e^{-n\rho/6}$. A union bound and (17) show that, with probability at least $1-\beta$, every consistent function in $\mathcal{H}_K$ has error at most $\rho/6$. On this event, the posterior-averaged test error is at most $\rho/6 + \rho/6 = \rho/3$. At any input whose correct-label probability is at most $2/3$, the posterior wrong-label probability is at least $1/3$. Integrating therefore bounds the mass of such inputs by $\rho$. □

**Corollary 11** (Fixed exponents above one)**.** *Fix either online construction, including its permitted anisotropy. Consider the independent-Gaussian prior*

$$\mathrm{Var}(W_{ai}) = \frac{\tau^2 N^{1-\gamma}}{d\lambda_i}, \qquad \mathrm{Var}(v_a) = \tau^2 N^{-\gamma-1}, \tag{18}$$

*with the same margin and exact-label conditioning.*

1. *For every fixed $\gamma > 1$, the unit-prefactor choice $\tau = 1$ has all the online conclusions at every sufficiently large width. The polynomial IID bound in Theorem 10 holds there as well.*
2. *For every prescribed online failure tolerance and IID accuracy parameters, one sufficiently small positive $\tau$ makes both conclusions hold for every $N \geq 1$ and every $\gamma \geq 1$ simultaneously.*

*The covariance asymmetry remains present. No conclusion here covers $\gamma = 1$ with unit prefactor $\tau = 1$.*

*Proof.* The mean-field scale is $t_N = \tau N^{(1-\gamma)/2}$. For fixed $\gamma > 1$ and $\tau = 1$, it tends to zero as $N \to \infty$. For $\gamma \geq 1$ and $N \geq 1$, it is at most $\tau$. Apply the width-uniform thresholds in Proposition 9 and Theorem 10. □

## 8 What exact isotropy changes

The strict energy ordering inside the majority tier is necessary for this proof. It cannot be inferred by choosing one preferred minimizer when the covariance is exactly isotropic.

**Proposition 12** (The isotropic tier)**.** *Set $\delta = 0$ and keep the same two-sided calibration. At every width, the minimum-energy consistent functions are exactly the $q$-coordinate majorities. The strong-prior posterior is uniform over whichever of them remain consistent. Under any continuation, its expected number of sampling mistakes before this tier is exhausted is at most*

$$1 + \log \binom{d}{q}.$$

*This bounds only this tier, not the entire later learning process.*

*Proof.* Lemmas 3 and 4 give the common candidate energy $\Phi_d(q)$ and outsider lower bound $\Phi_d(q+1)$. Coordinate permutations preserve the isotropic Gaussian law and the global-margin event, and act transitively on $q$-coordinate majorities. Thus their prior function probabilities are exactly equal at every positive variance, including all parameter-volume factors. Conditioning on exact labels preserves equal masses among the survivors. The outsider energy gap implies that the cold posterior is uniform on those survivors.

If a query eliminates a fraction $p < 1$ of a surviving class of size $m$, its sampling-error probability is $p$ and the decrease in its logarithmic size is $-\log(1-p) \geq p$. Summing gives at most $\log \binom{d}{q}$ before the final exhaustion, which costs at most one additional expected mistake. The same conditional argument applies to an adaptive continuation that does not observe the current posterior sample before choosing its query. □

An arbitrarily small positive $\delta$ splits the candidate energies and yields exponentially many certain cold-limit mistakes. At $\delta = 0$, the equal-mass mixture has the logarithmic bound above. The limits of zero covariance asymmetry and zero prior scale therefore need not commute. The theorem establishes the all-width lower bound for the stated perturbed covariance and leaves the corresponding exactly isotropic question open.

# A An explicit rigidity gap

We prove the numerical bound used in Theorem 7. This argument applies both to finite networks and to the probability measures in (15).

**Lemma 13** (Quantitative rigidity)**.** *Let $q \geq 3$ be odd, $d = 9q$, and $\Delta_q = 1/(10000q^3)$. If a globally weak-margin Boolean function has exactly $q$ essential input coordinates and a representation of isotropic energy at most $\Phi_d(q) + \Delta_q$, it is a signed majority on those coordinates. Every function with more than $q$ essential input coordinates has energy at least $\Phi_d(q) + 27/64$.*

*Proof.* Write the representation as in (15), and let $S$ be its essential support of size $q$. Put

$$K = I(\nu), \quad C = (\mathbb{E}_\nu b^2)^{1/2}, \quad u = \operatorname{arctanh}(1/(2C)), \quad t_0 = \tanh u = \frac{1}{2C}, \quad e = K - \Phi_d(q).$$

For each essential $j$, take an oppositely labeled adjacent pair, ordered by coordinate $x_j = +1$ and $-1$. Let $s_j(a)$ be the half-difference of its two hidden tanh outputs and let $\varepsilon_j \in \{-1, 1\}$ be its label orientation. Then

$$|s_j(a)| \leq \tanh |a_j|, \qquad \operatorname{sign}(s_j(a)) = \operatorname{sign}(a_j), \qquad \varepsilon_j \mathbb{E}_\nu[b s_j] \geq \frac{1}{2}.$$

With $r_j^2 = \mathbb{E}_\nu a_j^2$, the essential-coordinate argument gives $r_j^2 \geq u^2$ and the nonnegative decomposition

$$\begin{aligned} e = d \sum_{j \in S} (r_j^2 - u^2) + d\mathbb{E}_\nu \sum_{i \notin S} a_i^2 \\ + \left[ dqu^2 + C^2 - \Phi_d(q) \right]. \end{aligned}$$

The outside sum includes the bias coordinate. In particular,

$$r_j^2 - u^2 \leq e/d, \qquad \mathbb{E}_\nu \sum_{i \notin S} a_i^2 \leq e/d. \tag{19}$$

Using the test value $u_* = 1/\sqrt{6q}$ and $\coth t \leq 1/t + t$ for $0 < t \leq 1$ gives $\Phi_d(q) \leq 3q + 1$. Thus $e \leq \Delta_q < 1$ implies $K \leq 4q$, and

$$C \leq 2\sqrt{q}, \qquad \frac{1}{4\sqrt{q}} \leq u \leq \frac{2}{3\sqrt{q}} < \frac{1}{2}. \tag{20}$$

The lower bound uses $u \geq 1/(2C)$, and the upper one uses $dqu^2 \leq K$.

Put $h(s) = \tanh^2 \sqrt{s}$. Besides $0 < h'(s) \leq 1$, we have

$$-h''(s) \geq \frac{1}{5} \qquad (0 \leq s \leq 1). \tag{21}$$

Indeed, writing $t = \sqrt{s}$ and $T = \tanh t$ gives

$$-h''(s) = \frac{\operatorname{sech}^2 t \, [T - t + 3tT^2]}{2t^3} \geq \operatorname{sech}^2 t \, \frac{T^2}{t^2}.$$

For $0 < t \leq 1$, concavity of tanh and $3/4 < \tanh 1 < 4/5$ imply that the last expression is greater than $(9/25)(9/16) = 81/400 > 1/5$; the endpoint follows by continuity.

For $u \leq 1/2$, define the tangent deficit

$$D_u(s) = h(u^2) + h'(u^2)(s - u^2) - h(s).$$

For every real $a$,

$$D_u(a^2) \geq \frac{u^2}{10}(|a| - u)^2. \tag{22}$$

If $a^2 \leq 1$, (21) gives $D_u(a^2) \geq (a^2 - u^2)^2/10$, which implies the claim. If $a^2 \geq 1$, convexity of $D_u$ and (21) give $D_u(1) \geq 9/160$ and $D_u'(1) \geq 3/20$. Therefore $D_u(a^2) \geq (9/160)a^2$, which is stronger than (22) because $u^2 \leq 1/4$.

Cauchy–Schwarz gives $\mathbb{E}s_j^2 \geq t_0^2$, and $\mathbb{E}h(a_j^2) \geq \mathbb{E}s_j^2$. Using (19) and $h' \leq 1$ yields $\mathbb{E}D_u(a_j^2) \leq e/d$, hence

$$\mathbb{E}(|a_j| - u)^2 \leq \frac{10e}{du^2}. \tag{23}$$

Set $v = b/C$, so $\mathbb{E}v^2 = 1$. The oriented margin inequality gives

$$\|s_j - \varepsilon_j t_0 v\|_{L^2(\nu)}^2 \leq \mathbb{E}s_j^2 - t_0^2 \leq h(r_j^2) - h(u^2) \leq e/d, \tag{24}$$

$$\|s_j - \tanh(a_j)\|_{L^2(\nu)}^2 \leq \mathbb{E}[h(a_j^2) - s_j^2] \leq e/d. \tag{25}$$

The second line uses the common signs and pointwise magnitude ordering of $s_j$ and $\tanh(a_j)$.

Insert successively $u\,\mathrm{sign}(a_j)$, $t_0\,\mathrm{sign}(a_j)$, $\tanh(a_j)$, and $s_j$, and use $u/t_0 \leq 2$. Equations (23)–(25) give

$$\|a_j - \varepsilon_j uv\|_{L^2(\nu)} \leq 3\sqrt{\frac{10e}{du^2}} + 4\sqrt{\frac{e}{d}} \leq 12\sqrt{\frac{e}{du^2}} \leq 16\sqrt{e}.$$

Here (20) gives $du^2 \geq 9/16$. The outside coordinates have $L^2$ contribution at most $\sqrt{(d+1)e/d} \leq 2\sqrt{e}$ at every augmented cube input. Consequently,

$$\left\| a \cdot \bar{x} - uv \sum_{j \in S} \varepsilon_j x_j \right\|_{L^2(\nu)} \leq (16q + 2)\sqrt{e} \leq 18q\sqrt{e}.$$

By the Lipschitz property of tanh and Cauchy–Schwarz, the output is uniformly within $36q^{3/2}\sqrt{e}$ of

$$F\left(\sum_{j \in S} \varepsilon_j x_j\right), \qquad F(t) = C\mathbb{E}\big[v \tanh(uvt)\big].$$

For every nonzero $t$, the function $F(t)$ has the sign of $t$, since its integrand has that sign whenever $v \neq 0$ and $\mathbb{E}v^2 = 1$. The signed sums are nonzero because $q$ is odd. If $e \leq \Delta_q$, the uniform approximation error is at most $36/100 < 1/2$. A weakly margin-separated output cannot have the opposite sign at any input. This proves the first assertion.

For the second assertion, Lemma 3 gives the lower bound $\Phi_d(q+1)$. At its minimizer $u_{q+1}$,

$$\Phi_d(q+1) - \Phi_d(q) \geq du_{q+1}^2, \qquad u_{q+1}^2 \geq \frac{1}{4\Phi_d(q+1)}.$$

The test value $u^2 = 1/(2\sqrt{d(q+1)})$ and the same coth bound give $\Phi_d(q+1) \leq \sqrt{d(q+1)} + 1 \leq 4(q+1)$. Thus the gap is at least $9q/[16(q+1)] \geq 27/64$. □

Finally, $du_q^2 \leq \Phi_d(q)/q \leq 4$ gives

$$\Phi_d(q+s) \leq \Phi_d(q) + 4s \qquad (s \geq 0).$$

Under (13), every candidate in the linear-prefix construction has energy strictly below $\Phi_d(q) + 4\delta \leq \Phi_d(q) + \Delta_q/2$. Every outsider has energy at least $\Phi_d(q) + \Delta_q$, by Lemma 13; increasing coordinate weights cannot reduce that energy. This completes the explicit quantitative justification of Theorem 7.

**Verification scope.** The arguments are conventional mathematical proofs. They have undergone independent AI-agent mathematical reviews and are not Lean formalized. The source, reproducible build, and local audit records are included with this manuscript.

## References

[1] ChatGPT-6 Astra, Claude (Anthropic), Kaarel Hänni, and Dmitry Vaintrob. *Online learning of AND under Gaussian neural-network priors*. Research manuscript, 30 October 2026. Companion manuscript, section on an arbitrarily small anisotropy and its width-one majority corollary.